

UgvNet01 Model Information and AI Assistance Prompt

You are working with my custom deep learning model named **UgvNet01**.

This is not ResNet50, VGG16, or EfficientNet. This is my own custom PyTorch architecture designed for skin disease image classification.

Your task is to understand this model completely and help me improve, train, evaluate, deploy, and extend it.

Model Name

UgvNet01

Type:

- Custom PyTorch CNN Architecture
- Attention-based lightweight deep learning model
- Multi-class image classification model

Main Application:

- Skin Disease Detection
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Model Resources

PyPI Package

The model is available as a Python package:

Installation:

```
pip install ugvnet-model
```

Usage:

```
from ugvnet import predict

result = predict("skin_image.jpg")
```

```
print(result)
```

Hugging Face Model Weight

Pre-trained weight:

<https://huggingface.co/Onik-Howlader/UgvNet01>

Weight file:

UgvNet01_best_model.pth

Direct weight URL:

https://huggingface.co/Onik-Howlader/UgvNet01/resolve/main/UgvNet01_best_model.pth

GitHub Repository

Source code:

YOUR_GITHUB_LINK

The repository contains:

- Model architecture
 - Package structure
 - Setup files
 - Documentation
 - Usage examples
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Model Architecture

UgvNet01 consists of:

Input:

224 × 224 × 3 RGB Image

Architecture:

Input Image

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Stem Convolution

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UgvBlock 1

- Depthwise Convolution
- Pointwise Convolution
- Batch Normalization
- SiLU Activation
- CBAM Attention

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UgvBlock 2

- Depthwise Separable Convolution
- CBAM Attention

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UgvBlock 3

- Depthwise Separable Convolution
- CBAM Attention

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UgvBlock 4

- Depthwise Separable Convolution
- CBAM Attention

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Global Average Pooling

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Dropout

↓

Fully Connected Layer

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6 Disease Classes

Attention Module

UgvNet01 uses:

CBAM (Convolutional Block Attention Module)

Components:

1. Channel Attention
2. Spatial Attention

Purpose:

- Focus on important skin lesion regions
 - Improve feature extraction
 - Reduce unnecessary background information
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Disease Classes

The model predicts 6 diseases:

1. Atopic Dermatitis (AD)
 2. Contact Dermatitis (CD)
 3. Eczema (EC)
 4. Scabies (SC)
 5. Seborrheic Dermatitis (SD)
 6. Tinea Corporis (TC)
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Dataset Information

Dataset type:

Skin disease image dataset

Training approach:

Augmented images are used for training.

Recommended split:

Training:
80%

Validation:

10%

Testing:
10%

Image preprocessing:

Resize:
224×224

Normalization:
ImageNet mean/std

Current Research Goal

I want to create a thesis project using only UgvNet01.

Do not replace it with:

- ResNet50
- VGG16
- EfficientNet
- MobileNet

The objective is to evaluate and improve my own architecture.

Required Evaluation

Help me generate:

1. Accuracy
2. Precision
3. Recall
4. F1-score
5. Classification Report
6. Confusion Matrix
7. Training Accuracy Curve

8. Validation Accuracy Curve

9. Training Loss Curve

10. Validation Loss Curve

Training Configuration

Current framework:

- PyTorch
- Torchvision
- Google Colab

Expected training:

Optimizer:

AdamW

Loss:

CrossEntropyLoss

Learning Rate:

0.001

Epoch:

20-30

Early stopping:

Patience = 5

Possible Future Improvements

Analyze and suggest improvements for:

1. Model Architecture

Possible improvements:

- Better attention modules

- Additional feature fusion
- Residual improvements
- Multi-scale feature extraction
- Regularization techniques

2. Training Improvement

Suggest:

- Learning rate scheduling
- Data augmentation strategies
- Class balancing
- Label smoothing
- MixUp/CutMix
- Weight initialization

3. Deployment

Suggest ways to deploy:

- Web application
- Mobile application
- REST API
- FastAPI backend
- Flask application
- Docker deployment

4. Other Possible Applications

Although UgvNet01 was created for skin disease detection, it can be adapted for:

- Plant disease classification
- Animal disease image classification
- Medical image classification
- X-ray image classification
- MRI image classification
- Histopathology image classification
- Defect detection in manufacturing
- General image classification tasks

To use it for another domain:

Only these things need changing:

1. Dataset
2. Number of classes
3. Final classification layer
4. Training process

Instructions for AI Assistant

When helping me:

1. Understand UgvNet01 as my custom model.
2. Do not assume it is ResNet, VGG, or EfficientNet.
3. Provide PyTorch-based solutions.
4. Preserve my architecture unless improvement is scientifically justified.
5. Help me prepare thesis-quality experiments.
6. Help me create research papers and documentation.
7. Suggest improvements without replacing the entire model.

Always refer to this information before answering questions about UgvNet01.